Exploring the potential of coconut coir fiber as a biodegradable mulch: Properties, uses, processing methods, and applications

Author links open overlay panelChristian Lloyd Salva [a], Dinah Mae B. Ordista [a], Philip Donald C. Sanchez [a b], Joan Jane J. Sanchez [a b]

Show more

Share

Cite

https://doi.org/10.1016/j.biteb.2025.102377Get rights and content

Highlights

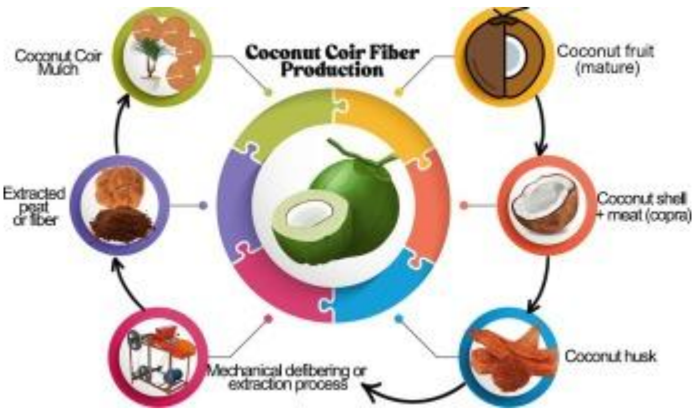
- Coconut coir is a natural fiber with low density and high toughness.
- Coir fiber traits highlight its potential as a biodegradable mulch.
- Coir improves soil protection, water retention, and weed control.
- Biodegradable mulches provide a sustainable option to plastic waste.

Abstract

Coir, a natural fiber derived from the husk of the coconut fruit and comprising approximately 10 to 17.5 % of the nut's total weight, has gained significant attention in recent years due to its availability, strength, and sustainability advantages. It is abundant, inexpensive, and possesses physicochemical and mechanical properties that make it favorable compared to plastic mulches, which have several drawbacks such as non-biodegradability, soil degradation, infiltration issues, and economic limitations. As a cellulosic material, coir is known for its high tensile strength (47.00–348 MPa), favorable density (0.85–1.44 g/cm^3), and length (6.00–8.00 in), making it comparable to other natural fibers. The utilization of biomass materials, such as coconut coir fibers, holds great promise for the cost-effective production of biodegradable mulches. This review provides an overview of the physical, chemical, and mechanical properties of coir and the processes involved to extract fibers from coconut husks, covering both traditional methods

(manual scraping and retting) and mechanical approaches (defibering and decorticating machines) for sustainable use. It also explores the benefits and agricultural applications of coir fibers, with emphasis on their potential as biodegradable mulches, along with current production techniques and the challenges associated with their use. The current review will pave the way for researchers, farmers, and environmentalists in identifying future research gaps, particularly regarding the feasibility of coconut coir as a biodegradable mulch.

Graphical abstract



1. Download: Download high-res image (159KB)
2. Download: Download full-size image

Introduction

The need for sustainable practices in the agricultural sector has grown recently, with a preference for using biodegradable materials to minimize environmental impacts. Mulching is one prominent technique that is becoming more popular in agriculture. Mulching provides advantages to agriculture, with its potential use in soil moisture retention, temperature regulation, weed growth control, promotion of microbial activity, and landscaping in terms of cost, aesthetics, and the environment (Iqbal et al., 2020). Furthermore, because mulching regulates soil temperature, it alters the soil's physical and chemical composition, promoting crop development and early harvest (Salama and Geyer, 2023). Mulches are materials applied to cover soil surfaces and can be derived from various raw sources. They are generally categorized as organic or synthetic. Organic mulches include straws, husks, grasses, cover crops (live mulches), sawdust, compost, and manures, while the most common synthetic mulch is plastic (polyethylene) (Iqbal et al., 2020; Ye et al., 2023). However, the use of plastic in agriculture contributes significantly to environmental pollution, posing risks to ecosystems and adversely affecting plant growth, soil, and water quality (Hachem et al., 2023; Lakhiar et al., 2024). Global plastic production has surged from over two million tons in 1950 to more than 450 million tons in 2019 (Ritchie et al., 2023). A review by Sa'adu and Farsang (2023) further highlighted the widespread contamination of agricultural lands by plastics, with the highest shares reported in Asia (60 %) and Europe (29 %). Similarly, Long et al. (2023) demonstrated a strong link between macroplastic remnants in fields and the intensity of plastic film mulch use. Huang et al. (2020) identified plastic film mulching as a major source of macroplastic in agricultural lands, prompting the introduction of biodegradable mulches as alternatives. Soylu and Kızıldeniz (2024) reported that biodegradable film mulch outperformed plastic mulch in maintaining soil health, water use efficiency, nutrient availability, and microbial activity, suggesting its potential to optimize crop and soil health. However, Tao et al. (2024) found that biodegradable film mulch can degrade into micro- and nanoplastics, releasing additives that contribute to soil pollution.

In the contemporary context, biodegradable mulch is gaining popularity with its known benefits as a sustainable approach in agriculture. Biodegradable mulches were mainly composed of organic materials which are wastes in agricultural production. Correspondingly, Coconut (*Cocos nucifera* L.) husks, comprising 35 % of the entire coconut, are regarded as waste, and in some places, coconut husks remain an environmental concern (Tooy et al., 2022). Despite being regarded as waste, coconut husks contain remarkable fibers called coir. Coir is a coarse, brown fiber that is resistant to saltwater and microbes, making it suitable for various applications involving soil and water (Mishra and Basu, 2020). A review by Stelte et al. (2022) highlighted the expanding applications of coir fibers derived from coconut husk. Traditionally used in products such as textiles, mats, and brushes, coir is now increasingly being explored as reinforcing fibers in binderless fiberboards, natural fiber composites, construction materials, solid biofuels, and even as an absorbent for heavy metals and toxic substances. More recently, natural fibers have also been applied in the production of biodegradable mulches. Liu et al. (2021) reported that biodegradable mulch made from natural fibers effectively regulates soil temperature, suppresses weed growth, reduces light transmittance, and demonstrates superior biodegradability compared to degradable plastic films. Given that coconut husks are predominantly composed of coir fibers, their potential as a sustainable mulching material is drawing increasing attention.

Hence, the demand for coco coir in recent years has dramatically increased for fabricating new environmentally safe and sustainable biocomposites (Hasan et al., 2021a). Studies have revealed their significant role in controlling slope erosion and enhancing soil and water holding capacity (Asha'ari et al., 2021). A study conducted by Aiome and De Silva (2014) utilized coir dust as a mulching material. The results revealed its feasibility as a mulching material due to several factors. Firstly, it exhibited high electrical conductivity, implying high moisture content, which is advantageous in dry conditions. Secondly, it demonstrated high water retention, further contributing to its suitability in water-stressed environments. Additionally, coir dust mulch exhibited a high composition of organic matter, promoting plant nutrition availability, productivity, and growth. Furthermore, it showed a low pH, which contributes to its resistance to biodegradation, as microorganisms struggle to survive in low-pH environments.

The demand for sustainable agriculture necessitates a greener solution. In the context of mulching, numerous studies have uncovered the adverse effects on the ecosystem of using plastic (polyethylene) mulch. It not only harms the environment but also affects the health of plants, soil, and water. Despite the number of studies supporting the negative impact of plastic on the environment, it is still being used in agricultural practices like mulching due to its efficiency and lower cost (Lakhiar et al., 2024). The utilization of coconut coir fiber as a biodegradable mulching material has the potential to address the issues associated with plastic mulch. However, despite its potential, relatively few studies have explored the feasibility of coir fiber as a mulching material. Ali et al. (2022) critically reviewed its utilization in cementitious materials, while Shukla et al. (2022) provided an overview of its applications in road construction. More recently, Bawono et al. (2025) discussed the use of coir fiber in fiber- and wood-based panels.

Several studies have been published regarding the development of fiber-based biodegradable mulches. However, there is a lack of recent research on fiber extraction, benefits, and potential utilization, specifically focusing on coconut coir fiber as a biodegradable mulch solution. For instance, Liu et al. (2021) utilized fiber-based degradable nonwoven mulch, mainly concentrating on natural fibers from recyclable mill waste. Marasovic et al. (2024) discussed biodegradable nonwoven mulches from natural and renewable sources, focusing on jute, hemp, viscose, and polylactide fibers. Asha'ari et al. (2021) evaluated the performance of using coconut husk and fiber as a practical solution to control slope erosion. Aiome and De Silva (2014) utilized coir dust as a mulching material. However, there has been limited exploration of the production or extraction of fibers from raw materials, which can vary among different commodities. Additionally, there are few mentions of the benefits and potential utilization of coconut coir fiber as a biodegradable mulch. Therefore, the main objective of this study is to provide a comprehensive overview of the extraction processes for coir fiber production from coconut husk, as well as the benefits, use, utilization, and potential applications of coconut coir fiber as a biodegradable mulch. Furthermore, this paper highlights the mechanical processes involved in coir production, with emphasis on decortication and defibering machines, the general characteristics of the fiber, and its potential applications in agriculture and construction, such as soil erosion control, animal bedding, biofuel, and biodegradable mulching technology. Overall, this review aims to assist the agricultural sector, particularly coconut farmers, in valorizing coconut husk waste or obtaining an organic mulching material for their local farms. In addition, this comprehensive review provides background information on the utilization of coconut coir fiber as a biodegradable mulch, which may serve as a valuable reference for future researchers interested in the topic.

Properties of coconut coir fiber

Coconuts (*Cocos nucifera* L.) are primarily cultivated for their oil-rich copra, leaving the husk as a major by-product. The copra meat is encased within a thick shell and husk, which consist mainly of coir and pith. Coconut husks, as shown in Fig. 1, accounting for about 35 % of the fruit, are often treated as waste despite being a rich source of coir fibers (Tooy et al., 2022; Bawono et al., 2025). Coir from mature coconuts is brown, thick, and tough, whereas coir from immature coconuts is

Benefits and uses of coconut coir fibers

Coir has been tested as an alternative to peat in horticulture and is currently the most widely used substitute in this sector (Schmilewski, 2008). The fiber obtained from the coconut husk, commonly known as coco coir, is gaining popularity as a natural growing medium for a variety of crops. Increasing pressure on producers, retailers, and growers, particularly in the horticulture sector, has driven demand for alternative, renewable, and reliable substrates (Ceglie et al., 2015). When selecting

Potential applications of coconut coir fibers

Coir fiber, derived from coconut husks, is recognized as one of the sturdiest natural fibers in commercial use. Its slow decomposition rate is a notable advantage for creating durable products (Hao et al., 2018). With the increasing demand for sustainable practices and growing concern over environmental constraints, exploring the potential utilization of coconut coir is crucial. It offers a sustainable solution as a biodegradable material, while also adding economic value for coconut farmers by

Overview of the biodegradable mulch

Biodegradable mulch is a type of mulching material designed to cover the soil surface, suppress weeds, conserve soil moisture, regulate soil temperature, and improve crop growth, like plastic mulches, but with the key difference that it naturally breaks down in the soil over time through the action of microorganisms (bacteria, fungi), sunlight, and environmental condition (Yang et al., 2020). Biodegradable mulch is a sustainable approach toward greener agricultural production. Such sustainable

Challenges and future prospects

The future of biodegradable agricultural mulches is marked by both promise and significant challenges. A primary obstacle remains their high cost, averaging 2.54 times that of conventional polyethylene mulches (Hayes et al., 2012; Selke et al., 2011), which limits adoption, especially among small-scale farmers. While the current market emphasizes sustainable agriculture, optimizing natural resource use and promoting environmentally friendly, organic farming presents an even tougher frontier,

Conclusion

This review highlights the important properties and expanding modern applications of coconut coir fiber, which is recognized as a natural biofiber with favorable characteristics and capable of effectively substituting synthetic fibers in polymer- and plastic-based mulching materials. Based on the review, coir fiber is a durable, tough, and resilient cellulosic material resistant to moisture, fungi, rot, and extreme temperatures. Its use extends beyond traditional handicrafts to applications

CRediT authorship contribution statement

Christian Lloyd Salva: Writing – original draft, Visualization, Methodology, Conceptualization. **Dinah Mae B. Ordista:** Writing – original draft, Visualization, Methodology, Conceptualization. **Philip Donald C. Sanchez:** Writing – review & editing, Visualization, Supervision, Conceptualization. **Joan Jane J. Sanchez:** Writing – review & editing, Validation, Supervision, Conceptualization.

Declaration of competing interest

The authors declare no conflict of interest, and the funders had no role in the design of the study; in the collection, analyses, or interpretation of data; in the writing of the manuscript, or in the decision to publish the results.

Acknowledgment

The authors are grateful to the Department of Agricultural & Biosystems Engineering, College of Engineering and Geosciences and the Center for Resource Assessment, Analytics, and Emerging Technologies (CSU-CReATe) for the facilities and technical expertise afforded during the entire conduct of the study. Special acknowledgment is given to the Department of Budget and Management (DBM) for funding the research project (RS - Valorization of Agricultural Waste into Potential Utilization) under the